Lagrangian driven Bezier Curve optimisation for robotics trajectory generation

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Abstract

We propose a method for robotic trajectory generation based on optimizing Bezier curves using the principle of least action. The method treats the trajectory as a particle path, with a Lagrangian composed of kinetic and potential terms representing velocity smoothness and obstacle penalties. Control points of the Bezier curve are optimized to minimize the corresponding action integral, yielding smooth and dynamically feasible trajectories. This approach reduces manual tuning compared to traditional spline-based methods.

1 Introduction

Trajectory generations have always been one of the most important aspects of robotics. Having well designed trajectories for specific tasks can have a notable impact on efficiency and cost control. However, it is often challenging to achieve the optimum trajectories. In most scenarios, splines will be used as robots must move in a continuous and dynamically feasible way due to it's physical constraints. But constructing such splines can be difficult, as ensuring the right amount of curvature and smoothness requires many trial and error. Constructing the spline itself is often difficult, and designing an optimal trajectory may also be challenging, as the first attempt is rarely optimal.

By using discrete control points to construct a smooth, continuous curve by means of a formula and optimizing the control points with according to the least action can significantly reduce testing times when creating such trajectories.

2 Bezier Curves

A Bezier curve is a parametric cruve defined by a set of control points P_0 through P_n, where n is the order of the curve (1 for linear, 2 for quadratic, 3 for cubic and so on). And the first and last control points being the end points.

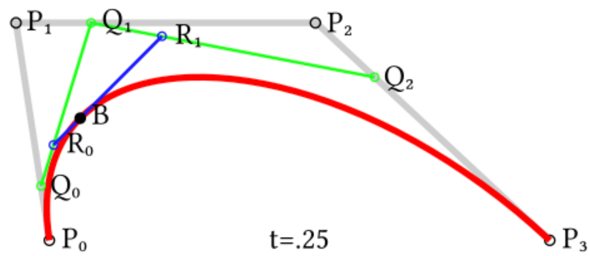

Figure 1: **Visualization of the construction of a high order Bezier curve**

The general explicit definition of a Bezier curve of any degree n is :

$$B(t) = \sum_{i=0}^{n} \binom{n}{i} (1-t)^{n-i} t_i P_i$$

and it's first order derivative being :

$$B'(t) = n \sum_{i=0}^{n-1} \binom{n}{i} (1-t)^{n-i} (t) (P_{i+1} - P_i)$$

3 The Principle Of Least Action

In classical mechanics, the principle of least action states that the actual path taken by a system is the extremum of it's action S, where the action of a system is defined by :

$$S[x^A(t)] = \int_{t_i}^{t_f} \mathcal{L}(x^A(t), \dot{x}^A(t)) dt$$

where $\mathcal{L}$ is the Lagrangian of the system, $\mathcal{L} = T(\dot{x}^A) - V(x^A)$.

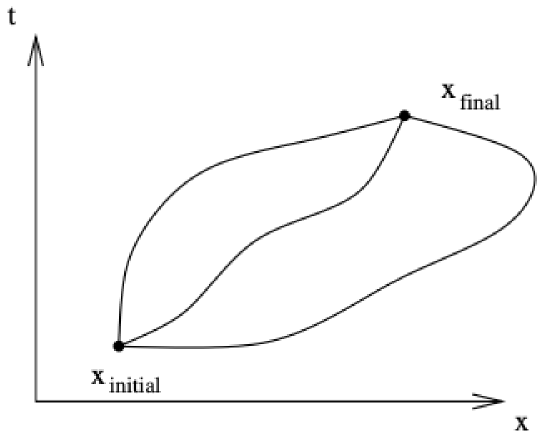

Figure 2: **Visualization of a particle's path with according to the least action.** Out of the family of possible trajectories connecting two points, the physical trajectory is the one that extremizes the action integral. In the case of a free particle, this corresponds to the path in the middle , which coincides with the shortest distance.

4 Application

By modifying the pre-existing trajectory generator: PedroPathing, we are able to perform optimization on the generated Bezier Curves by generating another new Bezier curve with the optimized control points. To achieve this, we constructed the lagrangian $\mathcal{L} = \frac{1}{2}[B'(t)]^2 + V(x)$, where the repulsive potential energy is

$$V(x) = \begin{cases} \infty, \text{ if } x \leq r \\ 0, \text{ if } x > r \end{cases}$$

. The action integral $\int_0^1 \mathcal{L}[x]dt$ and the control points are then fed into the Nelder-mead optimization algorithm as a cost function to be minimized.

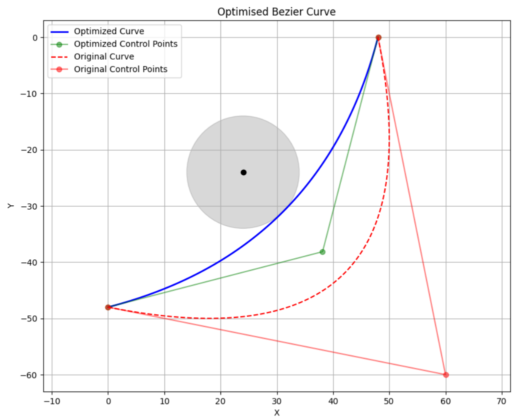

Figure 3: **Optimization result of a trajectory curving around an obstacle**

5 Conclusion

This work presents a novel approach to robotic trajectory generation by combining Bezier curves with the principle of least action. By formulating the trajectory optimization problem as minimizing an action integral with a Lagrangian that includes both velocity smoothness terms and obstacle avoidance penalties, we demonstrated that physically-motivated optimization can produce smoother trajectories with reduced manual tuning compared to traditional spline-based methods. The proposed method successfully generates continuous, dynamically feasible paths that naturally avoid obstacles while maintaining smoothness.

However, several significant limitations emerged from this investigation. The fundamental mismatch between Bezier curves, which were designed for computer graphics applications, and the requirements of optimal trajectory generation became apparent. While Bezier curves provide smooth interpolation, they cannot capture all possible optimal paths. Additionally, the computational burden presents a major obstacle for practical implementation. The exponential growth in required iterations as the number of control points and obstacles increases creates severe scaling issues. This computational complexity makes real-time applications impractical for complex environments.

Future work should explore alternative curve representations that are better suited to trajectory optimization while maintaining the benefits of the least action principle. Potential directions include investigating B-splines with optimization-friendly basis functions, developing approximation methods to reduce computational complexity, or hybrid approaches that combine the smoothness benefits of this method with more computationally efficient algorithms for real-time applications. Despite its current limitations, this work establishes a foundation for physics-based trajectory optimization that could inspire more practical solutions in robotics path planning.